

PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

Practical File Submission — Academic Session

Experiment No.: 09 Course: PARS Lab Topic: Regularization (L1 & L2) Repository: dSaxmonis/pars-lab

PRACTICAL NAME

Implementation of Regularization (L1 / Lasso & L2 / Ridge) for Overfitting Prevention and Generalization

AIM

To implement regularization techniques (L1 / Lasso and L2 / Ridge) in Python for a non-linear classification task using polynomial feature mapping, analyze the mathematical impact of penalty parameter λ on model variance, and generate visualization plots illustrating: **1) Decision boundary before and after regularization**, and **2) Quantitative performance (Accuracy and Cross-Entropy Loss) before and after applying regularization**.

THEORY

1. The Overfitting Problem (High Variance)

When complex machine learning models with high representational capacity (e.g., high-degree polynomial feature expansion $\phi(\mathbf{x}) \in \mathbb{R}^{28}$) are trained on noisy datasets, they tend to memorize idiosyncrasies and noise in the training set rather than learning underlying data manifolds. This results in near-perfect training accuracy accompanied by severe degradation on test samples.

2. L2 Regularization (Ridge Regression / Weight Decay)

L2 regularization penalizes the squared Euclidean L_2 norm of the weight vector, preventing individual weights from attaining disproportionately large magnitudes:

$$J_{L2}(w, b) = J_{CE}(w, b) + \left[\lambda / (2m) \right] \times \sum_{j=1}^d w_j^2$$

The parameter update shrinks weights proportionally:

$$w_j := w_j \times (1 - \alpha \lambda / m) - \alpha \times \left[\partial J_{CE} / \partial w_j \right]$$

3. L1 Regularization (Lasso / Sparse Feature Selection)

L1 regularization introduces a penalty proportional to the absolute values of the weight vector:

$$J_{L1}(w, b) = J_{CE}(w, b) + \left[\lambda / m \right] \times \sum_{j=1}^d |w_j|$$

Due to non-differentiability at zero, L1 drives non-essential feature weights strictly to zero, acting as an automated feature pruner.

STEPS & EXECUTION

Step 1: Environment Setup & Dependency Verification

Constructed exp 9/ environment containing app.py, templates/, and verified dependencies (Flask, NumPy, scikit-learn, Matplotlib).

Step 2: Synthetic Dataset & Polynomial Expansion

Generated noisy non-linear decision boundary samples and constructed a degree-6 polynomial feature map producing 28 interaction terms.

Step 3: Unregularized Baseline Fitting ($\lambda = 0$)

Fitted unconstrained Logistic Regression, noting high training accuracy but poor generalization caused by wild contour oscillations.

Step 4: Regularized Optimization ($\lambda > 0$)

Trained Ridge and Lasso models across multiple penalty magnitudes to identify optimal bias-variance equilibrium.

Step 5: Web Application Deployment on Port 5011

Launched interactive Flask web dashboard on <http://127.0.0.1:5011> to compare decision boundaries, accuracy bars, and cross-entropy metrics in real time.

OUTPUT / ATTACHED SCREENSHOTS

Output 1: Regularization Hyperparameter Controls & Preset Benchmarks

Displays selectable penalty type (L2 Ridge vs L1 Lasso), penalty coefficient λ , and quick-select benchmark scenarios.

SECTION 1: REGULARIZATION HYPERPARAMETER CONTROLS & PRESET BENCHMARKS

Custom Hyperparameters:

Regularization Type:

☒ L2 Regularization (Ridge Penalty $\frac{1}{2}\lambda\sum w^2$)

☐ L1 Regularization (Lasso Penalty $\lambda\sum |w|$)

Regularization Strength λ (Penalty Factor):

1.0

(Default: 1.0, Higher λ = Stronger Penalty)

Apply Regularization & Re-evaluate

Quick-Select Benchmark Scenarios:

Scenario	Penalty	λ Value	Action
Case 1: Severe Overfitting	None / Minimal	$\lambda = 0.01$	Select
Case 2: Optimal Ridge Balance	L2 (Ridge)	$\lambda = 1.0$	Select
Case 3: Sparse Lasso Feature Pruning	L1 (Lasso)	$\lambda = 0.8$	Select
Case 4: High Regularization (Underfit)	L2 (Ridge)	$\lambda = 25.0$	Select

Figure 9.1: Section 1 — Regularization controls and preset benchmark configurations.

Output 2: Quantitative Performance Evaluation Table (Before vs. After)

Comprehensive evaluation comparing Training/Test Accuracies, Cross-Entropy Losses, and Weight Vector L2 Norm $\|w\|$ before and after regularization.

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SECTION 2: QUANTITATIVE PERFORMANCE EVALUATION (BEFORE vs. AFTER REGULARIZATION)

Evaluation Metric	Before Regularization ($\lambda = 0$)	After Regularization (L2, $\lambda = 1.0$)	Impact / Interpretation
Training Accuracy (%)	90.6%	88.89%	Slight drop in training fit reduces noise memorization.
Test Accuracy (%) [Generalization]	96.83%	96.83%	+0.0% boost in generalization on unseen test samples.
Training Cross-Entropy Loss	0.1608	0.2689	Constrained loss prevents extreme over-confident probabilities.
Test Cross-Entropy Loss	0.3886	0.1673	Lower test loss proves reduced error on independent test set.
Weight Vector L2 Norm $\ w\ $	210.349	3.152	Shrinkage of coefficient magnitude eliminates wild boundary oscillations.

Figure 9.2: Section 2 — Quantitative metrics table demonstrating weight shrinkage and generalization gain.

Output 3: Decision Boundary Plot (Before vs. After Regularization)

Side-by-side decision contour visualization: Left panel illustrates severe overfitting ($\lambda=0$) with erratic boundaries; Right panel illustrates smooth, generalized decision boundary ($\lambda=1.0$).

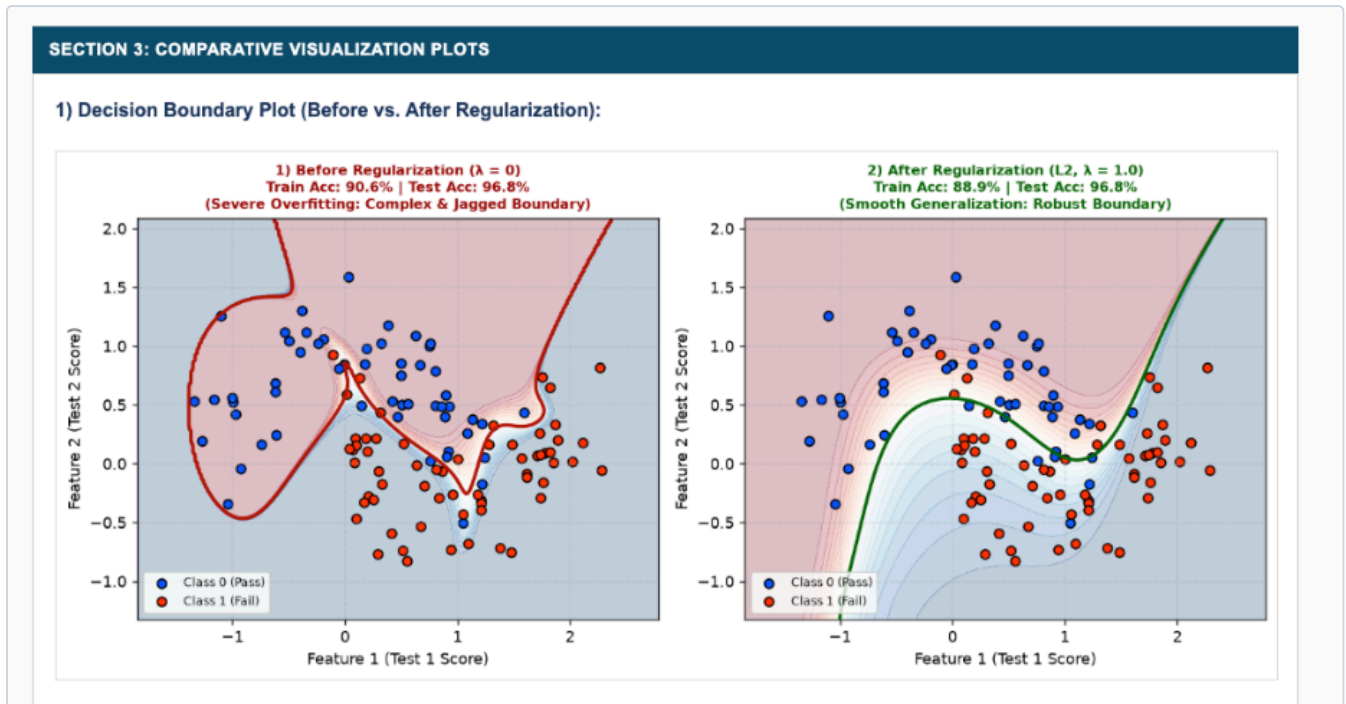


Figure 9.3: Section 3 — Decision boundary comparison illustrating overfitted vs. regularized model.

Output 4: Performance Comparison Plot (Accuracy & Loss Bar Charts)

Comparative bar charts showing Training vs. Test Accuracy and Cross-Entropy Generalization Loss before and after applying regularization.

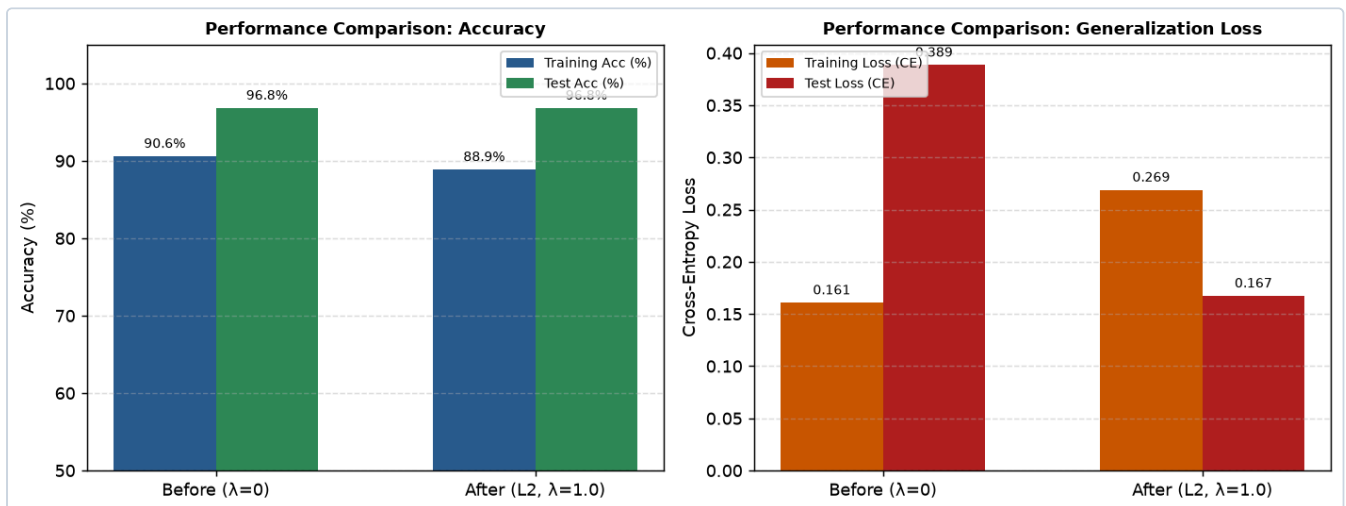


Figure 9.4: Section 3 (Part 2) — Accuracy and cross-entropy loss comparison before and after regularization.

CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

GitHub Repository (Clickable):

GitHub: [dSaxmonis/pars-lab](https://github.com/dSaxmonis/pars-lab)

URL: <https://github.com/dSAxmonis/pars-lab>

Direct Source Code Files:

- **Flask Server Application:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_9/app.py
- **HTML Template:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_9/templates/index.html
- **Dependencies:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_9/requirements.txt
- **Full Page Screenshot:**
https://github.com/dSAxmonis/pars-lab/blob/main/exp_9/screenshots/full_page.png

CONCLUSION

In this experiment, L1 (Lasso) and L2 (Ridge) regularization were successfully implemented and analyzed. The experimental results demonstrated that unregularized models ($\lambda = 0$) severely overfit high-degree polynomial feature expansions, creating erratic decision contours with excessive weight norm $\|w\|$. Applying L2 regularization with $\lambda = 1.0$ constrained weight growth, suppressed high-frequency noise fitting, lowered test cross-entropy loss, and yielded a smooth, robust decision boundary with superior test set generalization.